



Management of Potential Cervical Spine Injuries in Adults

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AUTHOR/S

This Clinical Guideline has been developed by Emergency Medicine unit in consultation with Neurosurgery, Diagnostic Imaging and Physiotherapy unit(s).

TARGET AUDIENCE and SETTING

This guideline applies to all Monash Health patients with suspected and confirmed cervical spine injury and the Monash Health Medical doctors, registered and enrolled nurses, midwives and allied health staff caring for such patients (within their scope of practice).

DEFINITIONS

- **Patient** - a person who is receiving care in a health service. The term patient encompasses all other relevant terms that may be used in different health care settings including patients, people/persons, families, carers, clients, residents and people with lived experience of specific areas of ill health.
- **Cervical collar** - Medical device used to stabilise/support and limit the movement of a person's cervical spine
- **Soft collar** – single use, flexible, fabric covered soft foam neck support that provides mild support to limit neck movement
- **Short-term foam collar** – rigid, foam collar designed to maintain neutral alignment of the cervical spine after injury to the neck. These collars have limited pressure relieving properties and are therefore only worn for a short period of time (up to 2-3 days). An example of this is the Philadelphia[®] collar
- **Long-term plastic collar** – rigid, plastic collar with soft, foam padding engineered to provide substantial motion restriction of the cervical spine. These collars have good pressure relieving properties and are suitable to be worn for prolonged periods (eg 6 months). An example of this is the Aspen Vista[®] or Miami J[®] collars.

BACKGROUND

Rationale for the guideline:

There is no scientific evidence that any type of cervical collar—whether used during pre-hospital transport or initial trauma management—is effective in stabilising an acutely injured cervical spine or in preventing further neurological deterioration in patients with spinal cord injury (SCI)¹. In contrast, there is increasing evidence that rigid collars can contribute to significant complications and morbidity when used to immobilise the cervical spine^{2,3}.

For example, a recent meta-analysis demonstrated increased mortality (RR=2.5, 95% CI 1.07, 5.41) associated with spinal immobilisation in patients with penetrating trauma, with no benefits observed with respect to reversible spinal injury⁵.

Reflecting this evidence base, Ambulance Victoria updated its clinical practice guidelines in 2021 to apply **soft collars**—rather than semi-rigid collars—for patients with suspected cervical spine injury.⁶

In alignment with Ambulance Victoria, all patients with suspected cervical spine injury at Monash Health are now managed with a soft collar. This approach does **not** indicate a downgrading of spinal care and does not mean that the cervical spine has been cleared. Rather, it is part of a holistic spinal care model that includes early clinical care, appropriate imaging, diagnostic review, and ongoing treatment planning for confirmed or suspected spinal injuries.

This change reflects a patient-centred, evidence-based approach endorsed by spinal and neurosurgical experts, and it prioritises both safety and patient comfort.

Patients presenting to Monash Health via ambulance will remain in a soft collar, which signals the need for further spinal assessment and imaging (if indicated) either in the Emergency Department (ED) or on the ward, depending on clinical progression. All patients with suspected cervical spine injury should be

maintained in a neutral position or position of comfort; conscious patients can usually maintain this themselves.

In cases of altered mental state—due to delirium, cognitive impairment, or intoxication—attempts to enforce spinal immobilisation may increase harm. In these situations, a senior clinical decision maker should be consulted. For unconscious patients, a soft collar should be applied with in-line or spinal transfer precautions.

STANDARD REQUIREMENTS

When undertaking any clinical interaction with a patient, staff are expected to:

- Perform routine hand hygiene. Refer to the [Hand Hygiene Procedure](#).
- Introduce yourself to the Patient and Carer/ Family if in attendance.
- Explain what you do and what you are there for
- Check patient identification. Refer to the [Patient Identification Procedure](#).
- Obtain consent as per the [Consent to Medical Treatment Procedure](#).
- Keep the patient/carers informed and involve them in decision making.
- Document interaction in the electronic medical record or health record using black pen; including date, time, signature and designation.
- Documentation of all procedures, ordering of medications and pathology, insertion of lines and devices will be done in the appropriate EMR fields where available.

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CLINICAL GUIDELINE

Presentation to Monash Health service:

Via ambulance:	- When person presents via ambulance, continue with ambulance Victoria guidelines and maintain soft collar. - Any spinal boards should be removed straight away.	
Self-presents:	Conscious:	- Clinical judgement in assessment and decision making needed. - Patient to remain in position of comfort. - It is not required to lie the patient flat. - Patients that are lying flat may require padding under the occiput to optimise neutral alignment of the cervical spine. Don't put a soft collar on unless they have a) neuro signs and symptoms on arrival (*escalate for prompt medical assessment and cervical immobilisation in position of comfort) OR b) would be more comfortable with collar. - The soft collar should not force the neck into any position. The soft collar should fit around the neck.
	Altered mental state due to delirium, cognitive impairment or intoxication:	- Do not force a patient to lie down Involve a senior clinician decision maker to decide if patient appropriate to maintain themselves in neutral position or if a soft collar is needed necessary.
	Unconscious:	Use soft collar and in-line or spinal transfer
Fall or suspected injury on bed-based unit:	Maintain neutral or position of comfort and apply soft collar as appropriate	

Patient assessment:

- [Advanced care directives/GOC](#) discussion. Apply common sense and discuss with patients or their medical decision maker (MDM). Consider if treatment in line with their care directives and are they appropriate for immobilisation.
- Consider analgesia
- Careful history to elicit symptoms

- Comprehensive neurological examination for signs of spinal cord injury. Inclusive of:
 - UL segment
 - LL segment
 - Sacral segments (ie saddle paraesthesia, PR if indicated and patient consents)
- Musculoskeletal examination
- Complete NEXUS criteria⁶ – See Table 1. The NEXUS criteria is a validated tool used to define a low-risk cohort that doesn't require imaging.
 - If any of the following criteria are present, proceed with cervical spine imaging to further evaluate for a possible injury
 - i. Midline cervical tenderness
 - ii. Altered mental state
 - iii. Focal neurological deficit
 - iv. Evidence of intoxication
 - v. Painful distraction injury
 - If NEXUS criteria is positive, seek senior medical review and/or use the Canadian C-spine Rule⁷ to determine if imaging is needed.

Table 1. NEXUS criteria. Cervical spine injury cannot be excluded if any criterion are present¹²

Midline cervical tenderness

Present if pain is elicited on palpation of the posterior cervical midline from the nuchal ridge to the prominence of the first thoracic vertebra, or if pain is reported on palpation of any cervical spinous process

Altered mental status

Glasgow Coma Scale ≤ 14
 Disorientation to time, place, person or events
 Inability to remember three objects at 5 minutes
 Delayed or inappropriate response to external stimuli

Focal neurologic deficit

Any patient-reported or examiner-elicited neurologic deficit

Evidence of intoxication

Recent history reported by the patient or an observer of intoxication or intoxicating ingestion
 Evidence of intoxication on physical examination, such as odour of alcohol, slurred speech, ataxia, dysmetria, or other cerebellar findings
 Behaviour consistent with intoxication
 Tests of bodily secretions are positive for drugs (including but not limited to alcohol) affecting mental alertness

Painful distracting injury

Any condition thought by the clinician to be producing pain sufficient to distract the patient from a cervical spine injury. Examples may include:

- any long bone fracture
- a significant visceral injury eg abdomino-pelvic injury
- a large laceration, degloving injury, or crush injury
- extensive burns
- any other injury producing acute functional impairment

Figure 1: Canadian C-spine Rule

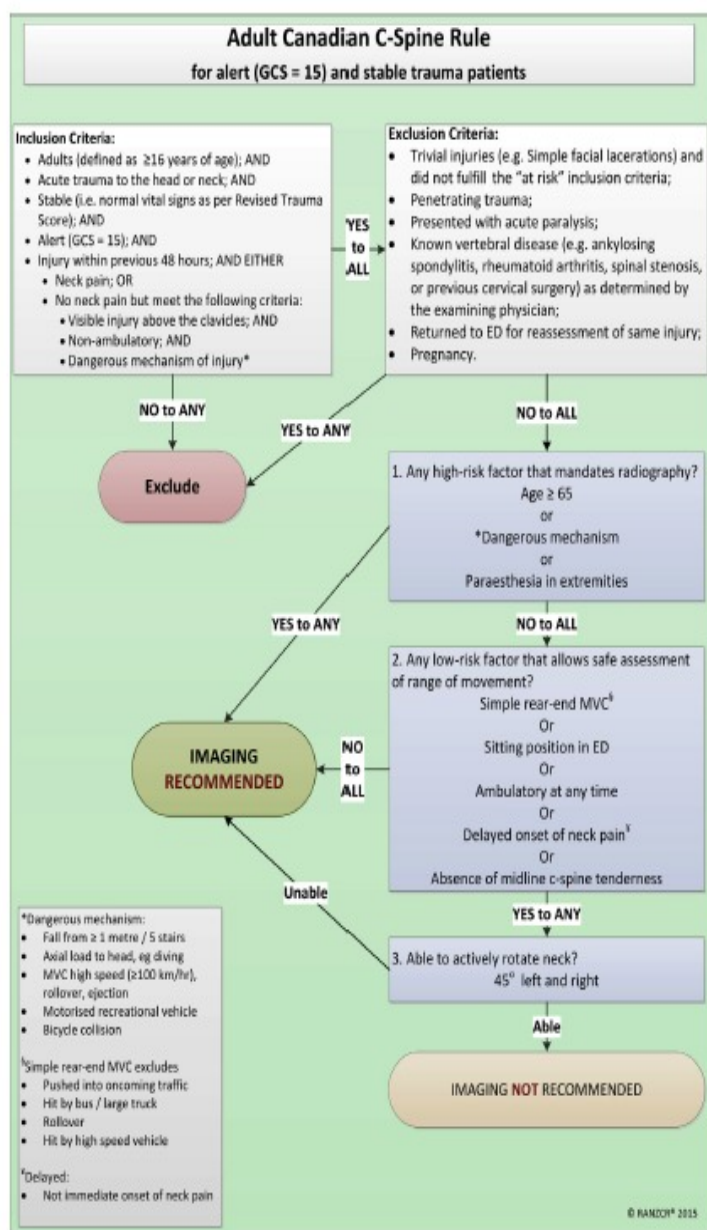
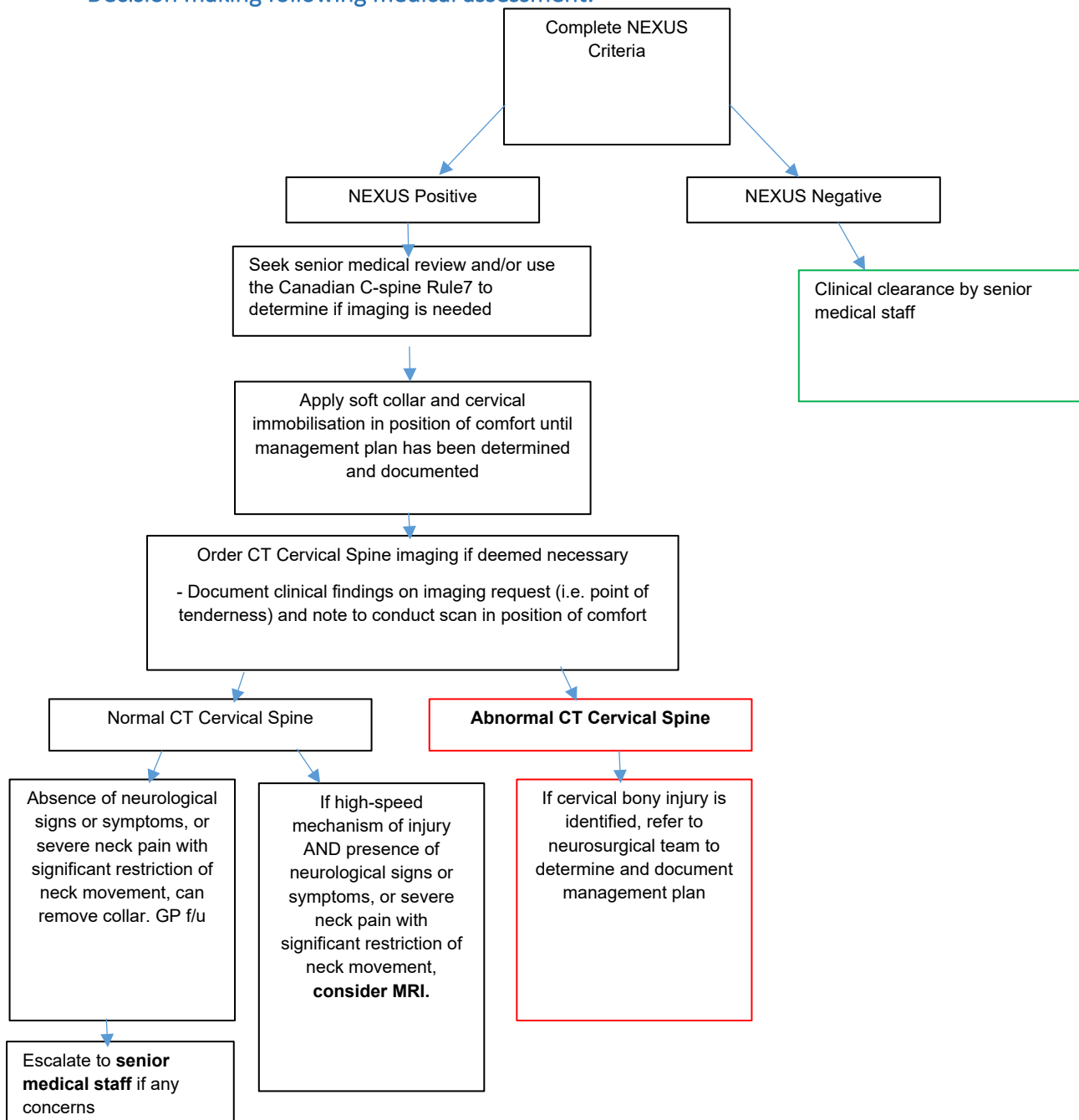


Fig 1. An example of a static PDF version of a decision support flowchart, illustrating the Canadian Adult C-Spine Rule (adult cervical spine trauma). ED = emergency department; MVC = motor vehicle collision.

Decision making following medical assessment:



- If NEXUS positive, apply soft collar and cervical spine immobilisation in position of comfort. Maintain soft collar and cervical spine precautions until a management plan has been determined and documented, and further diagnostics have been performed and/or reviewed.
- If NEXUS positive, use Canadian C-Spine Rule and discuss with senior clinician to determine if CT Cervical spine imaging is required. If ordering DI ensure:
 - Document clinical findings on imaging request, i.e. point of tenderness.
 - Note to have scan in position of comfort
- If normal CT Cervical Spine;
 - It is not advised to palpate for midline tenderness to guide practice at this point

- In absence of neurological signs or symptoms, or severe neck pain with significant restriction of neck movement – clear cervical spine
 - GP to follow-up and consider further imaging as required
 - NB: Escalate any concerns for discussion with senior medical staff.
- If normal CT Cervical Spine but high-speed/energy mechanism of injury AND ongoing presence of neurological signs or symptoms, or severe neck pain with significant restriction of neck movement, consider MRI.
 - In suspected cervical spine injury, MRI is used following CT to further assess patients deemed at risk of:
 - Disco-ligamentous cervical spine injury
 - Epidural haematoma
 - Spinal cord injury
 - Vascular injury such as vertebral artery dissection
 - If you suspect traumatic vascular injury, seek senior advice for investigation and management
- If CT cervical spine is abnormal (Acute injury or severe canal stenosis), refer to neurosurgery for determination and documentation of a management plan.
- Intubated and unconscious patients should be managed individually in discussion with Senior Medical Staff

Documentation:

- All assessment, imaging and management details should be clearly documented in the patient's medical record on EMR.
 - In situations where the decision not to treat/immobilise has been made, this should be clearly documented in EMR.
 - Medical staff can document assessment findings in EDtrauma note in EMR – and must include neurological assessment findings.
- If cervical injury is confirmed, neurosurgical team are required to document requirements for immobilisation.
 - STANDARD CERVICAL IMMOBILISATION
 - Which includes:
 - What: Cervical immobilisation in long-term hard cervical collar
 - When: Collar to be worn 24/24 (including showering)
 - Who: Full Ax2 spinal precautions for collar care
 - If the management differs to the above 'standard cervical immobilisation' recommendations, this needs to be clearly documented in the patient EMR notes using the 'Spinal Management Plan' note to guide physiotherapy, orthotists and nursing staff.

ADDITIONAL RESOURCES

Soft Collar Procedure

[Management of Potential Cervical Spine injuries under 16 year olds 16 Procedure](#)

[RACGP - Cervical spine – assessment following trauma](#)

REFERENCES

Monash Health wish to acknowledge the use of the following resources in the preparation of this procedure:

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- Alfred Health Cervical Spine management following traumatic injury procedure
- The Royal Melbourne Hospital Cervical Spine Management following traumatic injury procedure
- [Use of foam collars for cervical spine immobilisation - Initial management principles](#)

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KEYWORDS

Neck injury, cervical spine injury, collar, fall

Document governance	
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